Raza *et al.*
International Journal of Air-Conditioning and Refrigeration (2026) 34:2
https://doi.org/10.1007/s44189-025-00096-w

International Journal of
Air-Conditioning and Refrigeration

REVIEW Open Access

A CFD analysis of photovoltaic panel cooling: a review

Saleem Raza[1], Ahsan Farooq[2], Muhammad Sohail Shehzad[2], Wajahat Hussain[3], Zafar Abbas[4] and Ik-Tae Im[5*]

Abstract
Increases in operating temperature significantly degrade the efficiency of photovoltaic (PV) panels, leading to reduced power output and limiting the wider adoption of solar energy technologies. This paper reviews recent developments in PV cooling strategies, with particular emphasis on studies that employ Computational Fluid Dynamics (CFD) to analyze and enhance these approaches. The effectiveness of various active and passive cooling techniques—such as phase-change material (PCM) systems, water- and air-based cooling, and nanofluids—is compared. The review highlights that CFD has become a crucial tool for evaluating the influence of parameters such as coolant type, flow rate, film thickness, and environmental conditions on PV module temperature and performance. Key findings indicate that CFD-validated hybrid cooling strategies and the integration of advanced materials offer substantial potential for improving both efficiency and sustainability.

Keywords Photovoltaic (PV) panel, Computational fluid dynamics (CFD), Phase change material (PCM), Solar energy

1 Introduction

The depletion of fossil fuels and escalating environmental concerns have recently spurred a significant surge in interest in renewable energy sources. Solar energy, a plentiful and limitless resource, holds enormous potential for sustainable energy production. The ability of photovoltaic (PV) panels to capture solar energy is increasingly recognized as a key component in the global fight against climate change and in addressing the world's energy challenges.

However, PV panels, the fundamental technology for harnessing solar energy, come with several drawbacks, notably low efficiency and high temperature sensitivity. Solar panel efficiency is primarily affected by three elements: the operating temperature of the semiconductor cell, the quality of the semiconductor used, and the intensity of the solar radiation flux. While the fluctuations in solar radiation cannot be controlled, the issue of temperature can. As the panels heat up, the elevated temperature adversely affects the semiconductor properties, leading to a decrease in voltage output. Consequently, the overall energy conversion efficiency of the PV system is compromised [1]. Furthermore, prolonged exposure to high temperature can accelerate the degradation of the panel's materials, thereby reducing their operational lifespan and overall performance [1].

Given that PV panels are critical for meeting renewable energy targets aimed at combating climate change, especially accounting for a significant portion of solar energy

*Correspondence:
Ik-Tae Im
itim@jbnu.ac.kr
[1]Department of Mechanical Design Engineering, Graduate school, Jeonbuk National University, Jeonju, Republic of Korea
[2]Department of Mechanical Engineering, Mirpur University of Science & Technology, Mirpur, Pakistan
[3]School of Mechanical Engineering, Beijing Institute of Technology, Beijing, China
[4]Department of Mechanical System Engineering, Jeonbuk National University, Jeonju, Republic of Korea
[5]Division of Mechanical Design Engineering, College of Engineering, Jeonbuk National University, Jeonju, Republic of Korea

Springer

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

production, their susceptibility to high temperature poses a considerable challenge. This is particularly true in regions like the Middle East, where extreme heat leads to substantial energy losses. As the world transitions to renewable energy to reduce greenhouse gas emissions and combat climate change, enhancing the efficiency of PV systems becomes critical. Addressing these thermal performance issues aligns with the global push for sustainable energy solutions to achieve climate change goals, ensuring that solar power can reliably meet the growing demand for clean energy even under harsh environmental conditions.

To overcome these temperature-related issues and maintain optimal working conditions for PV panels, numerous cooling techniques have been developed. These techniques generally fall into two categories: active cooling, which employs mechanical systems (such as fans or liquid cooling) to enhance heat removal, and passive cooling, which relies on natural heat dissipation mechanisms (like convection and radiation). Hybrid systems combine both strategies to optimize cooling efficiency. Passive techniques are often preferred due to their simplicity and cost-effectiveness, as they do not require additional energy inputs [2].

Numerous studies have been conducted to address the aforementioned problems in PV panel. Air cooling techniques, for instance, have been experimentally and numerically investigated to improve the efficiency of PV cells across various applications and design conditions. Optimal air cooling conditions have also been determined [3]. Zhe et al. [4] performed a simulation analysis of PV panels both with and without water cooling system. Their study utilized a water cooling system as a heat exchanger and applied the system to a commercially available PV panel. They also examined the effect of the inlet water temperature on cooling performance. Vithanage et al. [5] compare the thermal performance of water-based nanofluids against regular water in a heat exchanger cooling setup for PV panels using computational simulations. The findings indicate that nanofluids containing metal oxide particles significantly enhance heat transfer compared to plain water, presenting them as promising alternatives for improving the efficiency of PV cooling systems. In 2024, Mahdi et al. [6] evaluated the use of ground source energy for cooling PV panels in Kirkuk, Iraq, through detailed CFD simulations in ANSYS Fluent. By analyzing a three-dimensional model over ten daytime hours, the results show that ground-source cooling reduced the panel temperature by about 28% compared to conventional methods and boosted power output by 6.5%, reaching up to 88 W. The mechanism involves ambient air absorbing heat from the PV panel and releasing it to the soil and bricks through underground pipes, leading to significant temperature reduction and improved PV efficiency. Antara et al. [7] used CFD simulations to compare four different channel shapes - circular, hexagonal, semi-circular, and square - in photovoltaic/thermal (PV/T) systems, all with equal cross-sectional area and flow rate. The results indicated that the circular channel provided the lowest PV cell temperature (317.95 K), highest electrical efficiency (14.70%), and highest thermal efficiency (44.18%). These advantages were attributed to the circular channel's faster water velocity and thinner boundary layer, which collectively enhanced heat transfer and overall cooling performance. Other study [8] investigated various hybrid PV/T collector designs using three-dimensional CFD simulations, supported by experimental validation with a box-type collector. Results showed that spiral flow heat exchangers provided the most uniform module surface temperature and minimized hot spots, while direct flow designs produced higher temperature gradients and hot zones. Significantly, adding cross-fins to the box-type collector considerably reduced these hot zones. Overall, the spiral and cross-finned box-type designs achieved the highest electrical efficiencies and heat transfer performance, making them the most effective configurations under different test conditions.

Although several review works [2, 9] have been carried out on experimental and analytical approaches, no comprehensive review has yet been performed on CFD-based PV cooling systems. CFD simulations offer unique advantages over experimental methods by effectively addressing challenges such as scalability, cost, and complexity. Unlike physical experiments, CFD allows for detailed analysis of thermal and fluid flow behaviors across various design configurations and climatic conditions, enabling optimization without the need for costly and time-consuming prototype testing. This review focuses on demonstrating how CFD can enhance the development of PV cooling systems by accurately predicting system performance and identifying critical parameters under varying environmental conditions. By comparing CFD results with experimental data, the study validates simulation models and highlights discrepancies, thereby revealing knowledge gaps in current research. It also provides guidelines for future studies, such as improving model accuracy, exploring advanced cooling techniques, and applying CFD to more complex and realistic scenarios. Overall, the review underscores the potential of CFD as a powerful tool to advance PV cooling technology, offering insights that experimental methods alone cannot achieve. Furthermore, the scope of this review includes a thorough analysis of the benefits and drawbacks of active and passive cooling techniques, emphasizing their effectiveness in improving PV performance under varied operating circumstances. Active and passive cooling techniques are compared across various

conditions to evaluate their efficacy in enhancing the performance of PV systems.

2 Active cooling techniques

Active cooling systems circulate a cooling medium using mechanical devices such as fans or pumps, thereby enhancing heat dissipation. Although these systems involve more complex construction due to the additional components required, their primary advantage lies in their superior heat removal capacity. Various working fluids are employed in active cooling systems, including air, water, thermal oils, and nanofluids. These solutions are appropriate when the electrical output of the solar panels exceeds the power consumption of the cooling system. Their ability to adapt to changing thermal conditions makes them essential components of modern thermal management strategies. Notably, active air-based cooling techniques can improve photovoltaic efficiency by approximately 5% to 10% [10]. The following subsections describe the primary active cooling techniques used for dissipating heat from PV cells.

2.1 Air cooling

Air cooling systems are commonly used in various devices for thermal management and temperature reduction. Although air is generally less effective than liquids as a coolant, it offers advantages such as minimal material usage and low operational cost [11]. Fig. 1 shows an air-cooled PV/T system. A study [12] has been made to improve the performance of PV modules by employing forced air cooling technique, and the results were compared with those from PV modules without any cooling. According to the findings, the use of air cooling increased the electrical efficiency by 7.2% and the power output by 6%.

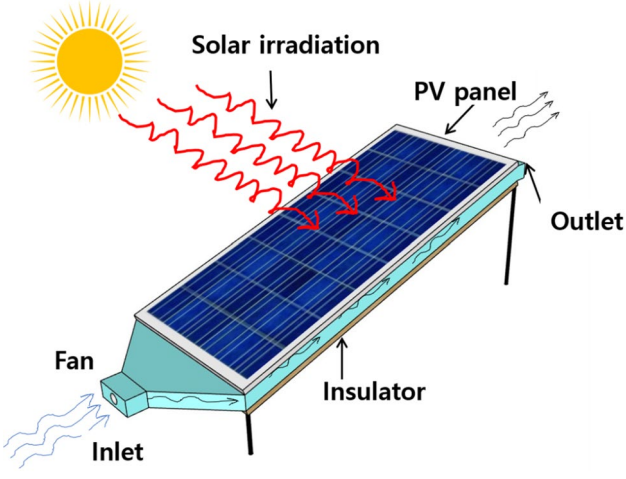


Fig. 1 Air cooled photovoltaic system

Due to fluctuating outdoor conditions, the efficiency of PV modules in real-world environments is generally lower than their rated module efficiency under standard test conditions (STC). Research on air-based PV/T systems has been further expanded, emphasizing various aspects of their design and performance [13, 14]. By effectively dissipating excess heat, air-cooled heat sink designs play a vital role in enhancing the efficiency and extending the lifespan of PV panels [15]. Utilizing ambient air for heat removal, air-cooled heat sinks provide a practical and cost-effective solution. The design of these heat sinks—particularly the number and placement of fins—is critical to maximizing thermal performance.

Hasan et al. [14] used CFD simulations to evaluate the effectiveness of various air-cooled heat sink designs for PV panel cooling. By varying the number and spacing of fins, they found that increasing the fin count significantly improved heat dissipation, with a maximum temperature drop of 38.44 K. Although finless or sparsely finned designs exhibited lower temperatures in certain regions, they resulted in less effective airflow. These findings highlight the critical role of fin number and spacing in optimizing heat sink design to improve thermal regulation, energy performance, and the longevity of PV panels. Patil et al. [15] conducted a CFD analysis of a solar PV/T system equipped with a bottom-mounted active air cooling mechanism. Using ANSYS Fluent software, they modeled an air duct with manifolds to ensure uniform airflow beneath the PV panel. The thermal behavior of the system was assessed at different air mass flow rates (0.04–0.1 kg/s) under 800 W/m² solar irradiance. At a flow rate of 0.01 kg/s, the maximum temperatures recorded on the top surface, PV cell, bottom surface, and air inlet/outlet were 48.8 °C, 48.4 °C, 47.3 °C, and 24.5 °C, respectively, demonstrating the effectiveness of the cooling system in controlling PV/T temperatures. In a subsequent study,

Sun and Shi [16] proposed an active air cooling system for a concentrating PV/T system equipped with a compound parabolic concentrator. Their study reported that the electrical, thermal, and overall efficiencies of the hybrid PV/T system increased with both the system length and the air mass flow rate. Wodołazski et al. [17] developed and analyzed a fully transient three-dimensional numerical model of a photovoltaic–thermoelectric generator (PV–TEG) hybrid system integrated with a single-phase inverter, using co-simulation in ANSYS. The study examined the effects of wind speed, solar radiation, and ambient temperature on system performance, considering thermoelectric phenomena such as Thomson, Seebeck, and Joule heating within the TEG. The combined use of a passive heat sinks and forced air circulation enhanced heat transfer, contributing to improved electrical efficiency. Özakin and Kaya [18] analyzed the energy and exergy performance of an air-based PV/T

system using ANSYS Fluent, comparing panel surface temperature distributions and airflow velocities within the cooling channel against experimental data. The system was tested under three fin configurations: no fins, sparse fins, and frequent fins. Results indicated that sparse and frequent fin configurations improved exergy efficiency by 70% and 30% for polycrystalline and monocrystalline panels, respectively, and thermal efficiency by 55% and 70%, respectively. However, the study did not consider the energy consumption of the cooling fan, which contributes to the system's overall energy demand and reduces the net energy gain from the PV panel. Furthermore, fan-based cooling becomes less effective under high ambient temperatures, where already-warm intake air limits cooling potential and may increase maintenance needs. These limitations suggest that air-cooling can be less reliable for consistently reducing elevated PV module temperatures.

In 2023, Liu et al. [19] explored the use of OpenFOAM for efficient and customizable CFD analysis of data center cooling and air distribution. The researchers developed a Java-based system for automated case setup and simulation, incorporating custom momentum sources to control airflow through servers. Validation against an Alibaba Cloud data center demonstrated the method's accuracy in predicting air temperatures across cold and hot aisles. This study provides a foundation for automated and cost-effective CFD simulations in data center design using open-source tools.

Several selected researches using air-cooling for PV systems are summarized in Table 1. Despite the benefits, PV/T air systems face three major challenges: (1) the requirement for fan power, which can offset the energy savings achieved through improved panel efficiency; (2) extreme weather conditions that reduce cooling effectiveness; and (3) the low heat capacity of air, which results in inferior thermal performance compared to other coolants such as water [20].

Table 1 Selected researches on air-cooled PV system

Reference	Year	Cooling method	Key findings
Tonui and Tripanagnostopoulos [11]	2007	Air channel flow with metal sheet and fins arrangements	• A theoretical model is developed and validated against experimental data. • Channel design
Sajjad et al. [12]	2019	Cooled air form Air-conditioning unit	• An economical PV module cooling method is presented. • PV module provided with cooling showed 7.2% higher electrical efficiency.
Ömeroğlu [13]	2018	Forced air circulation with copper fins	• Enhanced thermal efficiency by increasing airflow and fin count. • Electrical efficiency improved to 12.02% under high air velocity conditions.
Hasan et al. [14]	2024	Air flow with copper fins	• Heat sinks with a higher number of fins improve heat dissipation, with the 11-fin configuration demonstrating the highest temperature drop of 38.44 K
Patil et al. [15]	2024	Forced air flow	• PV cell arrangement is considered including top glass, EVA, solar cells and tedlar.
Wodołazski et al. [17]	2021	Forced air circulation	• Accounted the phenomena of Thomson, Seebeck, and Joule's heat place on the TEG system
Özakin and Kaya [18]	2019	Air flow with fins	• Sparse and frequent fin configurations improved exergy efficiency by 70% and 30% for polycrystalline and monocrystalline panels, thermal efficiency by 55% and 70%, respectively.

2.2 Water cooling

Water cooling is another active cooling technique used to dissipate excess heat from PV cells. It involves channeling liquid flow along the backside of PV modules to induce forced convection, as illustrated in Fig. 2. In 2020, the effect of front-face water cooling on PV panel efficiency and power output was studied using both CFD and experimental methods. The study [21] involved spraying water on the front surface of the PV panel at varying flow rates (8, 6, and 4 L per minute) through 4 mm nozzles. The results demonstrated an efficiency improvement of up to 16.2% at an ambient temperature of 45 °C compared to a conventional (non-cooled) panel operating under the same conditions. Vaishnav et al. [22] present the design and numerical analysis of a liquid-based, finned cooling system for solar panels that utilizes the heat generated by the panel as a medium to maintain optimal operating temperatures. A CAD model was developed in SolidWorks, and simulations were conducted using ANSYS Fluent to analyze flow patterns, heat transfer, and turbulence characteristics within the system. The study examines variations in temperature, pressure, and kinetic energy to assess the cooling system's performance under different conditions, thereby laying a foundation for the development of efficient thermal management solutions for PV systems. Antara et al. [7] conducted a CFD study evaluating the thermal and electrical performance of a PV/T system cooled using four different channel designs: circular, hexagonal, semi-circular, and square—all with the same cross-sectional area and mass flow rate. The results showed that the circular channel achieved the lowest PV cell temperature (317.95 K) and the highest electrical (14.70%) and thermal (44.18%) efficiencies among the four channel designs due to faster water velocity, a thinner boundary layer, and enhanced heat transfer rate.

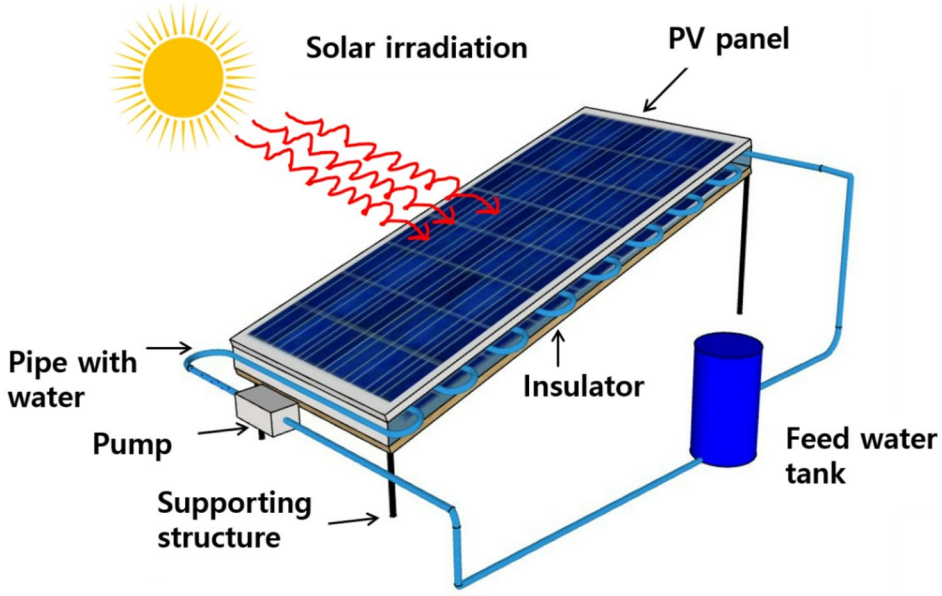


Fig. 2 A schematic of water cooling method for photovoltaic system

Dedić-Jandrek and Penga [23] provide an overview of various cooling strategies aimed at reducing the high operating temperatures of PV panels, which can reach up to 60 °C and result in efficiency losses of approximately 0.5% per degree Celsius. Their study focuses on active cooling through water spraying, which offers multiple advantages, including panel cooling, surface cleaning, and reduced infrared radiation absorption due to a lower refraction index. A computational fluid dynamics model was developed to simulate heat and mass transfer under both normal operating conditions and water spray cooling, showing strong agreement with experimental data. The study underscores the effectiveness of water spray cooling for PV panels and proposes potential directions for future research. Wu et al. [24] used CFD analysis to evaluate the thermal management of monocrystalline PV cells via water cooling. The study explored the effects of solar irradiance, ambient temperature, and wind speed on PV cell temperature and efficiency. The findings revealed that water cooling is particularly effective at maintaining lower cell temperatures under conditions of high solar irradiance and elevated ambient temperatures, thus preventing efficiency losses. While wind speed has a minimal effect when water cooling is employed, its influence becomes more significant in the absence of cooling. The study reported up to a 52% increase in cell efficiency compared to uncooled conditions.

Coolant replacement with liquids of higher thermal conductivity, such as ethylene glycol, has also been explored. However, the use of such fluids is often impractical due to high maintenance costs and operational complexity [25]. Channel geometry is one of the most influential factors in the heat transfer process [26]. Nowzari [27] investigated the performance of PV panels by analyzing their cooling behavior and heat flux. In his study, water was channeled above or below the PV module through a thin layer to prevent evaporation and minimize wastage. Six different models were evaluated using commercial CFD software (ANSYS), with aluminum fins incorporated in two of the designs. The results indicated that as the Reynolds number increased, the coolant outlet temperature decreased, while the heat flux increased. Building upon Wu's findings [24] regarding the benefits of water cooling for PV efficiency, Nowzari's investigation [27] further emphasizes the importance of optimizing the design and material of cooling channels and fins. This optimization is essential for effective heat dissipation and maintaining coolant temperatures, thereby addressing key challenges in practical thermal management. Future research should focus on efficient, stable, and controlled heat harvesting from PV panel surfaces. Additionally, the design, positioning, and material properties of fins can significantly influence heat transfer rates. Studies on water cooling system using CFD are summarized in Table 2.

2.3 Nanofluid cooling

Nanofluids absorb a significant amount of heat due to their excellent thermal conductivity. They can drastically reduce the surface temperature of PV modules and act as effective coolants, bringing PV systems closer to their optimal operating temperatures and thereby increasing power generation, as shown in Fig. 3. Furthermore, the heat absorption capability of nanofluids can enhance the

Table 2 Summary of selected researches on water-cooled PV system

Reference	Year	Cooling method	Key findings
Vaishnav et al. [22]	2021	Immersed fins in carbon-disulfide	• Efficient solar panel cooling system using fins and coolant.
Antara et al. [7]	2024	Water flow in four different types of channels: circular, hexagonal, semi-circular, and square.	• Circular channel gives the lowest PV cell temperature, 317.95 K, with an electrical efficiency of 14.70% and a thermal efficiency of 44.18%.
Wu et al. [24]	2021	Water circulation cooling	• Influence of solar irradiance, ambient temperature and speed of wind on the efficiency and temperature of a PV cell. • Maximum enhancement in the cell efficiency is approximately 52% in comparison with the case without applying cooling.
Kazem et al. [26]	2020	Water circulation in a channel	• Three channel models for water flow: direct, web and spiral f low. • The spiral flow system shows the lowest temperature for the PV module of about 45.2 °C and the highest PVT electrical efficiency of 9.1%.
Nowzari [27]	2019	Water flow in different designs of cooling channel	• The highest amounts of heat extracted from the panel is obtained in the case when the water flows over the module in a channel with a height of 3 mm.

performance of hybrid PV/T devices. References and their cooling method, key findings are summarized in Table 3.

A study by Aswad et al. [28] employing a three-dimensional CFD simulation model found that a serpentine circular pipe used as a heat exchanger could improve PV cell efficiency by approximately 10%. The addition of Al_2O_3 nanoparticles to water further increase the efficiency by 12%. These results underscore the importance of effective cooling techniques in hot climates, with nanofluids serving as promising coolants. The simulation outcomes were validated by experimental data using nanofluids as coolants in PV/T systems [29]. The mean absolute percentage error (MAPE) was within 10%, indicating good agreement between simulation and experimental results. Shrivastava et al. [30] analyzed the impact of a serpentine copper tube heat exchanger using nanofluids in 100 W PV/T collectors under Indian summer conditions. The cooling fluids tested were distilled water, 0.1% Al_2O_3, and 0.2% Al_2O_3 nanofluids. The study found that distilled water exhibited the highest reversibility, while thermal efficiency increased from 20% to 36% with nanofluids. Results also indicated that Al_2O_3 nanofluids, particularly at a 0.2% concentration, significantly enhanced thermal efficiency (up to 71.02%) and exergy efficiency (from 20% to 36%), while reducing irreversibility compared to water cooling. However, the analysis was limited to a single geographic and climatic condition, restricting the generalizability of the findings to other regions. Their group [33] also used CFD simulations to examine temperature distributions and cooling effectiveness across varying mass flow rates, corresponding to Reynolds numbers between 900 and 1,300.

In 2022, Hasan et al. [31] explored the use of nanofluids containing metal oxide nanoparticles in concentrated PV/T collectors equipped with a Fresnel lens. Five cooling fluids were tested: four types of nanofluids and pure water, for dissipating heat from multi-junction solar cells. The Nusselt number for all nanofluids was higher than that for water. Among them, water–SiO_2 nanofluid achieved the greatest heat transfer enhancement (12.5%), while water–Al_2O_3, water–ZnO, and water–CuO also showed notable improvements. However, the study did not address long-term stability, potential nanoparticle agglomeration, or maintenance concerns, all of which may affect practical deployment. Additionally, the environmental feasibility of large-scale implementation was not evaluated. They also presented a CFD simulation investigating the use of various metal oxide nanofluids as coolants in a concentrated PV/T (CPVT) system with multi-junction solar cells and a Fresnel lens. Nanofluids were circulated through mini-channels beneath the solar cells to evaluate heat transfer enhancement and surface temperature reduction relative to water. Results demonstrated that all nanofluids improved heat transfer, as indicated by higher Nusselt numbers with increasing Reynolds numbers. Water–SiO_2 nanofluid provided the greatest enhancement (12.5% over water) and reduced the top surface temperature of the mini-channel by 7%. This enhanced thermal dissipation led to improved electrical efficiency of the solar cells.

In the meantime, Omri et al. [32] proposed a novel cooling channel system incorporating porous deflectors (PDs) and nanofluids for PV thermal management. The system employed alumina nanoparticles dispersed in water as the cooling medium. Cooling performance was influenced by Reynolds number, Darcy number, PD number, and PD aspect ratio. The optimal configuration, with five PDs and an aspect ratio of 1, achieved a 107.5% increase in the average Nusselt number and a 13.7 °C reduction in temperature. This system demonstrated strong potential for PV thermal regulation. However, the study relied heavily on numerical simulations, which may not fully capture real-world complexities such as fluctuating environmental conditions and material behavior over time.

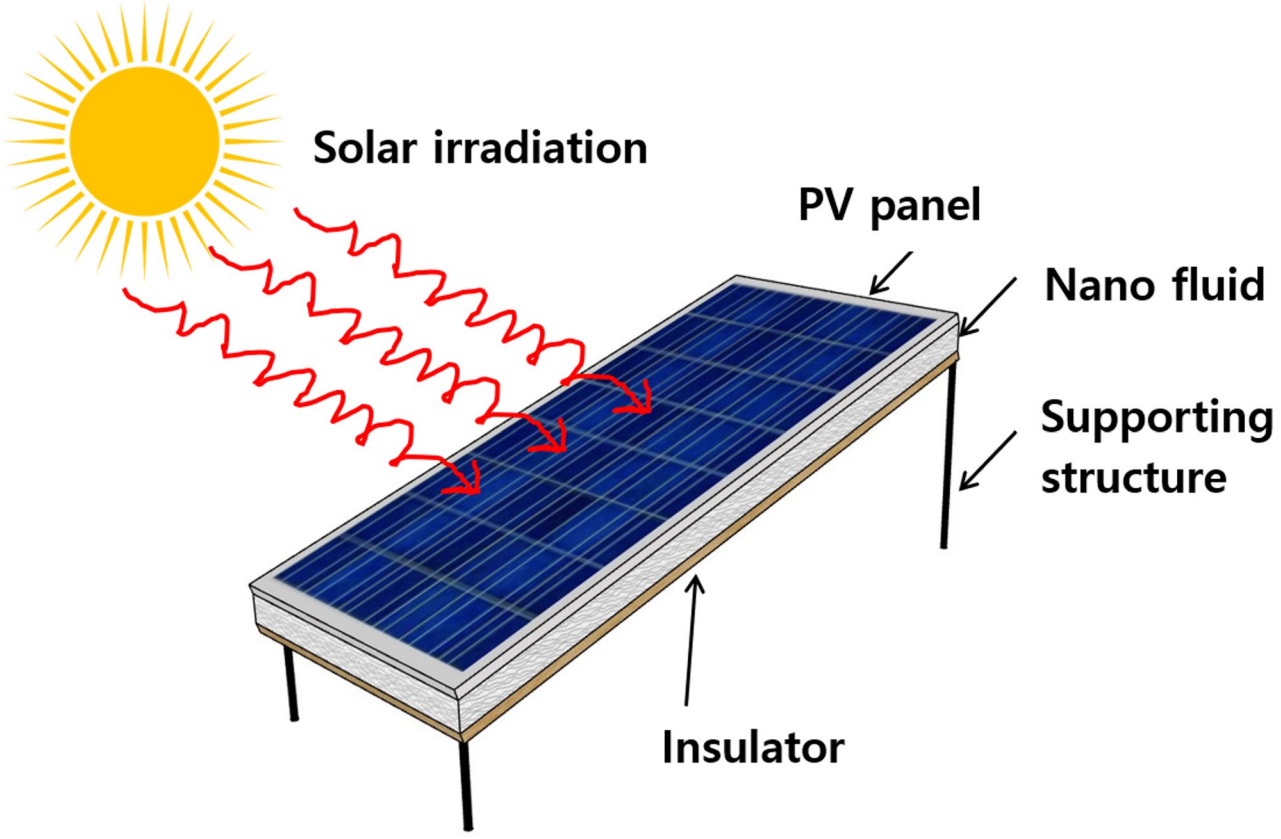


Fig. 3 Schematic of PV/T system with nanofluid as a coolant

3 Passive cooling techniques

In addition to active cooling techniques, passive cooling techniques are also highly suitable for cooling PV modules. Passive cooling does not require an additional power source to operate. Due to this advantage, these techniques have a simple structure and lower maintenance costs compared with active cooling methods. When phase change material (PCM) is connected to the back of a solar panel, it is expected to keep the solar cells at a lower temperature while providing higher efficiency. The subsections below explain how different passive approaches enhance PV cell efficiency.

3.1 Phase change materials

Employing phase change materials (PCM) to prevent high solar cell temperatures is one of the passive approaches for cooling PV cells, as shown in Fig. 4. Some strategies are used to improve PCM performance in lowering PV cell temperature. Recent advancements in PV cooling have been reviewed by Ali [34] through the integration of PCM-based systems. According to his review, PCMs are increasingly used for PV cooling because of their high thermal conductivity. Combining passive and active cooling solutions can further reduce PV cell temperature, increasing efficiency and thermal power output. Water-based hybrid PV/T systems can improve by 32% when combined with PCM, while nanofluid-based PV/T systems have been shown to increase efficiency by more than 60% when combined with PCM. Table 4 summarizes the recent CFD studies for PV cooling system using phase change material.

Nouira and Sammouda [35] analyzed the effect of integrating different PCMs of varying thicknesses, focusing on the importance of connecting PV panels with an appropriate macro-encapsulated PCM layer for improved thermal control. The study also examined the melting and solidification processes of specific PCMs. To model heat and mass transport, it is essential to investigate how wind direction, wind speed, and dust deposition affect PV panel performance. Dust deposition density reduces electrical power by 1.2, 2.8, and 3 Watts, respectively. Increased wind speed causes higher heat losses and lower panel temperatures. Critical limitations of this study include the melting and solidification cycles of PCMs, which may degrade over time, leading to reduced long-term performance. Environmental factors such as wind speed, direction, and dust accumulation present additional challenges.

Hossam et al. [36] utilized three-dimensional CFD simulations with ANSYS Fluent to analyze the thermal

Table 3 Summary of selected researches on PV cooling system with nanofluid

Reference	Year	Cooling method	Key findings
Aswad et al. [28]	2021	Alumina nanoparticles suspended water circulation through serpentine pipe	• Improvement of approximately 10% in the photovoltaic electrical efficiency when using the water cooling method and about 12% when using the nanofluid.
Rosli et al. [29]	2021	Alumina water nanofluid circulation in a serpentine tube.	• Numerical simulation results were validated using experimental data.
Jose et al. [30]	2023	Alumina water nanofluid circulation in a serpentine copper tube.	• Exergy and thermal efficiency were analyzed. • Good agreement between CFD and experimental results.
Hasan et al. [31]	2022	SiO_2, Al_2O_3, CuO, and ZnO nanoparticles in water channel flow	• The CFD results demonstrated that the Nusselt number of all nanofluids is greater than that of water. • Water-SiO_2 nanofluid has a significant impact on the heat dissipation.
Omri et al. [32]	2022	Multiple porous deflectors (PDs) and nanofluids in a channel.	• The combined utilization of PDs and nanofluid in the cooling channel provides an excellent tool for the thermal management of PV. • The best cooling performance achieved is 107.5% higher average Nu and 13.7 °C lower temperature as compared to reference case.

behavior and phase change dynamics in PV modules enhanced with PCMs. By modeling single and multilayer PCM configurations under hot climate conditions, the CFD analysis revealed that multilayer PCMs effectively lowered PV surface temperatures by up to 6 °C. The simulations provided detailed transient insights into heat dissipation and phase transitions, demonstrating the significant cooling potential of PCM integration for improving PV efficiency and performance in hot environments. Aneli et al. [37] investigated the performance of a conventional PV panel with two organic PCMs (PV-PCM) to reduce cell temperature rise and increase electrical performance. Unsteady numerical simulations were conducted using ANSYS Fluent software, evaluating PV cell temperatures, power production, and PCM thermal behavior. The study found that PV-PCM units achieved higher performance compared with conventional modules, especially during the hottest months, with an increase in peak power of 10% and 3.5% more energy produced annually. The study suggested that extending the simulation time for two days can better account for the degree of PCM solidification achieved overnight. One key limitation is that PCMs with low melting temperatures may not solidify completely at night, reducing their ability to absorb and store heat during successive daylight cycles. Thermal fatigue can also reduce PCM efficiency over time, requiring careful material selection and operational settings adapted to local environments.

Groulx et al. [38] studied the melting of PCM inside a rectangular enclosure, with and without fins, at various tilt angles. The study focused on temperature control of a finned PV panel filled with PCM. Four geometries

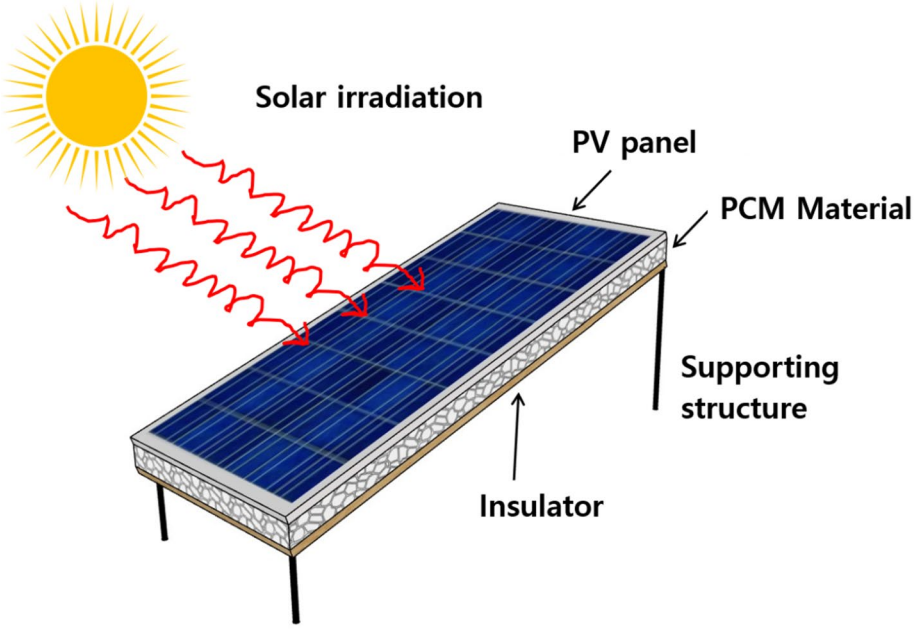


Fig. 4 A schematic diagram of PCM cooling method

Table 4 CFD analysis studies for PV cooling system using phase change material

Reference	Year	Cooling method	Key findings
Nouira and Sammouda [35]	2018	PCM layer is attached to PV cell.	• Exterior operating conditions such as wind direction, wind speed and dust accumulation on PV panel performance were analyzed. • 9 g/m^2 of dust deposition reduces PV panel power output of about 3 W at midday.
Hossam et al. [36]	2025	Single or multi-layer PCM container is attached to PV cells.	• The multilayered PCM arrangement lowering the temperatures up to 6 °C when compared with the single layered module coupled with PCM. • Enhancement in electrical efficiency of 1.6% and power output of 2.54% in comparison with conventional PV units.
Aneli et al. [37]	2021	PCM (RT28 and RT35) containing plates were attached to PV cells.	• The PV-PCM units achieved higher performances in comparison with a conventional PV module, especially during the hottest months. • An increase in the peak power of 10% and of 3.5% of the energy produced.
Groulx et al. [38]	2020	Finned PV panel filled with PCM (RT25) and installed at different tilt angles.	• Natural convection plays the major role in speeding up the PCM melting. • Optimized choice of the fin positioning was achieved in cases of "Full Fin" and "Front Fin" configurations.
Sumayli et al. [39]	2023	Thin tank filled with combination of PCMs with porous fillers.	• Copper foam with a porosity of 0.705: the highest thermal efficiency of 66.70% is achieved • Copper foam with a porosity of 0.94: the maximum electrical efficiency of 13.24% is resulted.
Khodadadi and Sheikholeslami [40]	2022	PCM with different finned containers.	• Increasing fin quantity improves reduction of temperature of PV panel. • Adding nano-powders to the PCM enhances the melting rate of PCM.

were considered: non-finned PCM enclosure, full-width fin, half-width fin, and inclined fin. Results showed that the most efficient thermal management was achieved when the PCM enclosure had a full-width fin attached to both the front and back plates. While the use of fins enhanced heat transfer through natural convection, it also increased material usage, weight, and manufacturing complexity, which could raise costs and complicate installation. Palm wax has also been used as a PCM to enhance passive cooling with a finned container heat sink [41]. The research compared three different PCM containers: grooved, tubed, and finned. The finned container showed the best cooling effect, reducing PV module temperature and achieving the highest electrical power output. The finned container also improved module efficiency from 9.33% to 9.82% (a 5.3% improvement) and the average performance ratio from 0.63 to 0.66 (a 4.8% enhancement). However, cooling performance strongly depends on solar irradiance, with significant benefits observed only when irradiance exceeds 500 W/m². This makes the solution less effective or unnecessary in regions with consistently low irradiance, limiting its universal applicability.

Sumayli et al. [39] used CFD to assess how copper metal foam with different porosities, incorporated into the PCM and collector of a PV/T system, enhances heat transfer and thermal regulation. Findings showed that metal foam improved thermal efficiency by facilitating PCM heat dissipation without notably reducing electrical efficiency, highlighting the effectiveness of PCM–metal foam composites for thermal management in PV/T systems. In 2022, Khodadadi and Sheikholeslami [40] simulated and evaluated a PV/T unit combined with paraffin in multiple finned containers, analyzing fin utilization, quantities, and configurations. Results showed that increasing the number of fins improved heat transfer, melting rate, and liquid fraction. Although adding nanopowders to paraffin improved the liquid fraction, it also raised issues regarding dispersion stability and potential accumulation.

3.2 Nano-PCM

For decades, the heat transfer capabilities of PCMs have been enhanced through nanotechnology. Nada et al. [42] employed RT55 paraffin wax for the thermal management of PV cells. A second thermal control approach involved adding alumina nanoparticles to pure PCM at a 2% concentration. Incorporating nanoparticles into PCM reduced the temperature of PV panels by 8.1–10.6 °C compared to those without a cooling device. These reductions improved the cooling system performance by 5.7% and 13.2%, respectively. The improvement is attributed to the increased thermal conductivity provided by the nanoparticles. Kumar et al. [43] conducted experimental studies on the effects of copper, silicon, and calcium carbonate on the thermal behavior and efficiency of PV panels. Two prototype designs were developed and tested: Prototype 1 consisted of a frame, PV panel, and electrical circuit, while Prototype 2 included an aluminum container attached to the PV plate containing the composite PCM. The results showed that the electrical performance of PV panels improved by an average of 2.8% with conventional PCM and by 4.3% with the composite PCM. However, the experiments were performed under controlled conditions, which may not fully capture real-world variables such as fluctuating solar irradiation,

wind speed, and dust deposition—all of which substantially influence PV performance.

To address the low thermal conductivity of PCM during PV cooling, Luo et al. [44] proposed the addition of expandable graphite to RT28. This modification produced a highly conductive composite PCM (7.571 W/m-K). The PV panel temperature decreased by 25 °C and remained low for an extended period. The maximum enhancement in power output achieved was 11.5%. In 2024, Seto et al. [45] compared the performance of a 50-Watt peak PV system with and without PCM and NePCM. Results showed that NePCM-ZnO performed better in reducing the operating temperature of the PV system, enhancing maximum PV output power by 10.85 W and maximum efficiency by 3.08%. These findings suggest that NePCM-ZnO can optimize the operational performance of PV systems. Nevertheless, the integration of nanoparticle-enhanced phase change materials (NePCM) faces challenges such as increased costs, uncertain long-term stability, and limited applicability under diverse environmental conditions.

A variety of heat transfer enhancement techniques—including the use of nanoparticles, porous metal foams, and extended surfaces—have been investigated for thermal energy storage in flat-plate solar collectors [46]. Configurations incorporating, I-shaped, Y-shaped, and T-shaped fins as extended surfaces were examined. The results demonstrated that each technique enhanced heat transfer performance, reducing melting time by up to 85.16% compared to pure PCM. Combining all techniques increased the heat storage rate by an order of magnitude. However, employing multiple methods simultaneously requires careful design to ensure compatibility and to avoid excessive pressure drops or flow resistance in the system.

4 Applications of CFD in PV cooling

Computational Fluid Dynamics (CFD) is a numerical simulation approach used to analyze and predict fluid flow, heat transfer, and other physical processes by solving the governing equations of fluid dynamics and thermodynamics, such as the Navier–Stokes equations. CFD employs discretization of the physical domain into small computational cells, enabling accurate representation of complex geometries and flow behaviors. This makes it a powerful tool for providing insights into fluid–surface interactions under diverse conditions.

In photovoltaic (PV) systems, CFD is an effective method for designing and optimizing cooling solutions. It solves the complex equations governing fluid motion, heat transfer, and energy conservation, producing detailed simulations of how different cooling techniques interact with PV panel thermal dynamics. This allows researchers and engineers to evaluate cooling strategies under various environmental conditions without the need for costly or time-consuming experiments. Moreover, CFD provides insights into flow patterns, vortex formation, and phase-change behavior in cooling materials. These capabilities support performance prediction, design optimization before prototyping, and the development of climate-specific solutions, ultimately improving the efficiency and reliability of PV systems in real-world applications. CFD has been widely used to assess and improve different PV cooling methods, and it is essential for extending the lifespan and efficiency of PV systems.

4.1 Passive cooling

CFD simulations have significantly improved the design of passive cooling solutions, enhancing heat dissipation by analyzing the dynamics of airflow around PV panels. Sheikh et al. [47] explores a novel approach using multi-layered PCMs to improve PV panel efficiency. The study found that a 15:85 thickness ratio between MPCM (metallic phase change material) and OPCM (organic phase change material) significantly reduced temperature rise, boosting performance by 35.8% and extending the operational life of PV installations. Applying adiabatic boundary conditions to the top and bottom walls simplifies the model. However, in real-world situations where heat exchange with the environment occurs, this assumption may not be valid. Alnakeeb et al. [48] investigated the performance of a hybrid PV–thermal with phase change material (PVT-PCM) system using ANSYS Fluent for detailed numerical analysis, finding that overall energy efficiency increases as the aspect ratio decreases. The upper layer of the PVT-PCM unit assumes homogeneous incoming radiation, which neglects the variability of solar radiation caused by atmospheric conditions, shading, and angle of incidence in real-world environments. As a result, thermal behavior estimations may fail to represent system performance adequately under varying solar conditions.

Kim and Nam [49] examined passive cooling technology in the PV market using a CFD simulation model. Attaching fins to the rear surface of the module effectively lowered the temperature, resulting in improved electrical efficiency. However, the methodology does not account for the impact of transmission materials such as glass and EVA (ethylene vinyl acetate) on solar irradiance transmissivity. The omission of these materials may lead to major errors in energy production projections, as they significantly influence the thermal characteristics and light absorption of the PV module.

As mentioned before, Nouira and Sammouda [35] studied the integration of different PCMs and macro-encapsulated PCM layers for improved thermal regulation. Their analysis showed that wind direction, speed, and dust accumulation affect panel performance, reducing

electrical power output and increasing temperature. Due to the strong influence of atmospheric conditions, shading, and angle of incidence on thermal dynamics, fluctuations in irradiance can result in uneven solar energy absorption on PV panels. In increasingly complex models, varying irradiance should be considered to provide a more realistic representation of energy absorption. Researchers must incorporate temperature-dependent characteristics, dynamic boundary conditions, and realistic solar radiation models to balance computational feasibility and accuracy when advancing the use of CFD in PV cooling.

4.2 Active cooling

Active cooling in PV systems uses fluids such as water or nanofluids to improve heat dissipation and maintain optimal temperatures. CFD helps engineers visualize flow distribution patterns, evaluate temperature reduction, and optimize cooling strategies.

Zhe et al. [4] conducted a study to forecast the temperature distribution of PV panels with and without a water cooling system using ANSYS, which enabled the investigation of the thermal model. The results reveal that panels without cooling have a larger average temperature distribution, while a water cooling system can lower it. The study uses environmental data from Perlis, Malaysia, but seasonal variations may affect its applicability. Wu et al. [24] explored the impact of water cooling on the efficiency of a monocrystalline PV cell using CFD. The findings indicate that water cooling greatly increases cell efficiency, particularly at higher ambient temperatures and solar irradiances. The study's simplified solar model ignores variations in the actual heat flow of PV modules. For more realistic thermal simulations, a dynamic model that considers these fluctuations is required. Furthermore, the study underestimates heat transfer rates and neglects convection on the upper surface of PV modules. In another study, Pomares-Hernández et al. [50] implemented simulations using COMSOL Multiphysics software for active cooling with water. However, many environmental factors that could greatly affect cooling performance, such as wind speed, ambient temperature, and solar radiation, appear not to have been considered in the models.

In 2022, Beldjani et al. [51] suggested using an air-cooled heat dissipator to enhance the cooling of PV panels, reducing heat generation and improving efficiency. Numerical simulations show that increased flow velocity enhances heat transfer. This study assumed constant solar radiation on a summer day, however, atmospheric conditions and shading cause variations in solar irradiance. Laminar flow, despite assuming turbulent flow, can offer benefits such as lower pressure drops and reduced energy consumption. As mentioned before, Patil et al. [15] analyzed a solar PV/T system with a bottom active air cooling system using ANSYS software. The study revealed that high operating temperatures reduce output power under the same solar radiation conditions. The analysis assumed a constant solar radiation of 500 W/m^2, but actual conditions, including time and location, can affect performance, necessitating a dynamic solar model.

As discussed in Sect. 3.2, the addition of nanoparticles improves the thermal conductivity and heat transfer properties of the fluid, making it more effective at removing heat from the PV system. Aswad et al. [28] examined the impact of temperature on PV cell performance using a three-dimensional CFD model. A serpentine circular pipe was used as a heat exchanger to transfer heat from the cell to the fluid. Alumina nanoparticles (Al_2O_3) were added to water at a 4% rate to improve thermal performance and efficiency. Results show a 10% improvement in PV electrical efficiency with water cooling and 12% with nanofluid, indicating the potential for increased efficiency in hot environments. The simulation's assumptions, such as treating viscosity and thermal conductivity as constant and assuming negligible radiation losses, limit its ability to accurately model complex physical phenomena. These simplifications can significantly affect heat transfer predictions and ultimately influence the overall thermal performance of the system.

Active cooling techniques represent a significant advancement in optimizing PV performance across diverse environmental conditions. By harnessing the power of CFD simulations, we gain deeper insights into these systems, paving the way for innovative cooling solutions that promise substantial improvements in energy efficiency and operational stability. Embracing these advancements not only enhances the viability of solar energy but also moves us closer to a sustainable energy future.

4.3 Hybrid cooling

Various methods have been studied and reported to cool PV panels. Therefore, research is being conducted to apply two or more of the various methods at the same time. Applying two or more methods, not only one method, can be called a hybrid cooling method. Hybrid cooling systems have demonstrated encouraging outcomes in enhancing solar panel efficiency by combining more than two methods. Computational fluid dynamics models have predicted significant drops in surface temperatures, highlighting the importance of heat transfer processes and airflow dynamics. Pomares-Hernández et al. [50] presented computational modeling of hybrid cooling. A mathematical model was developed to calculate output power changes resulting from temperature reduction. Results showed improved performance using hybrid methods. It is important to note that, despite the

Table 5 Commercial software used for CFD analysis for PV cooling

Reference	Year	Cooling method	Softwares
Zhe et al. [4]	2019	Water flow in the cooling case	ANSYS CFX
Patil et al. [15]	2024	Forced air flow	ANSYS Fluent
Wu et al. [24]	2021	Water flow in the panel attached o the back of the cell	ANSYS CFX 19.0
Aswad et al. [28]	2021	Alumina nanoparticles suspended water circulation through serpentine pipe	COMSOL Multiphysics
Nouira and Sammouda [35]	2018	PCM layer attached to PV cell.	COMSOL Multiphysics
Hossam et al. [36]	2025	Single or multi-layer PCM container is attached to PV cells.	ANSYS Fluent 19 R3
Aneli et al. [37]	2021	PCM (RT28 and RT35) containing plates were attached to PV cells.	ANSYS Fluent
Groulx et al. [38]	2020	Finned PV panel filled with PCM (RT25) and installed at different tilt angles.	COMSOL Multiphysics
Sumayli et al. [39]	2023	Thin tank filled with combination of PCMs with porous fillers.	ANSYS Fluent
Nemat-pourKeshteli [46]	2024	Nano-particles embedded PCM(RT82), porous metal foams and fins	ANSYS
Sheikh et al. [47]	2024	Multi-layered PCMs (Metalic PCM-Organic PCM)	ANSYS FLUENT 2021 R1
Alnakeeb et al. [48]	2024	Water flow in tubes embedded inside PCM (RT35)	ANSYS Fluent 17.42
Kim and Nam [49]	2019	Fins attached at the bottom of the PV module	Star-CCM+
Pomares-Hernández et al. [50]	2021	Water spray	COMSOL Multiphysics
Beldjani et al. [51]	2022	Fins in air flow	ANSYS Fluent
Birjandi et al. [52]	2021	Channel flow of nanofluid (MWCNT-Fe3O4/water)	ANSYS CFX 16.0
Nasef et al. [54]	2019	Direct contact PCM (RT35) for passive cooling system and water circulation system PCM in a tank for active system	ANSYS Fluent 17.2

model's assumption of turbulent flow, laminar flow may have advantages in certain situations, such as lower pressure drops and reduced energy consumption. Investigating laminar flow scenarios may provide insights into optimal design arrangements for various operating conditions, particularly at lower mass flow rates.

Birjandi et al. [52] investigated the use of a hybrid nanofluid as a coolant in PV cells, revealing that it can improve efficiency by 35.66% compared to water. However, the assumption that the thermophysical properties of coolants remain constant despite temperature variations may not hold true in real-world scenarios where significant temperature gradients can occur. This simplification could lead to inaccuracies in predicting thermal behavior over time. Potgieter et al. [53] used computational fluid dynamics and experimental validation to evaluate a unique solar air heater design for thermal efficiency and temperature distributions. The models over-predicted thermal efficiency by 6.75% to 9.01%. The simulation employed an HP 1000 weather station for realism and dynamic modeling, recording wind speed, temperature, and pressure. Nasef et al. [54] employed hybrid cooling by integrating a phase change material heat storage unit with a closed-loop water cooling system. This innovative approach achieved a remarkable 60% reduction in the average temperature of the CPV system. The study's analysis of CPV performance under different conditions could be improved by incorporating real-world testing.

CFD is increasingly used in hybrid cooling methods for PV systems, enhancing thermal management and energy efficiency. It simulates interactions between cells, fluids, and the environment, improving solar panel performance.

5 Concluding remarks

5.1 Validation and challenges

From published studies, it can be seen that various commercial software is used for the analysis of PV cooling. Table 5 summarizes the commercial software used in the CFD analysis study of PV cooling. The accuracy of simulations in PV cooling applications depends on the validation of CFD results against experimental data. Typically, this process involves comparing simulation outputs—such as temperature and heat transfer rates—with actual measurements obtained under experimental conditions. Most studies suggest that deviations should fall within the range of 5–10%, enabling researchers and engineers to make well-informed decisions regarding the design and optimization of PV cooling systems using CFD.

Although CFD has proven effective for optimizing PV cooling systems, challenges remain, including high computational costs, radiative heat transfer modeling, and mesh dependency. High-fidelity CFD modeling requires substantial computational resources, often necessitating a trade-off between accuracy and processing time as well as advanced hardware. Simulating radiative heat transfer continues to be a major challenge, potentially leading to inaccuracies in thermal predictions. Moreover, mesh dependency is a critical factor in CFD, as poorly designed or insufficiently refined meshes can yield unreliable results. Achieving the right balance between detail and practicality therefore remains a demanding task.

5.2 Future directions

Future research should focus on the following points:

- Enhancing the modeling of environmental interactions by accounting for variations in solar radiation, changes in wind patterns, and seasonal fluctuations in ambient temperature.
- Integrating machine learning with CFD to automate design optimization for PV cooling. Such integration enables the application of adaptive design techniques that respond to dynamic environmental conditions.
- Conducting field testing and validation to bridge the gap between theoretical predictions and practical applications, thereby ensuring that CFD remains a reliable tool for engineers designing real-world PV systems.
- Coupling CFD with energy simulation tools to evaluate the impact of cooling on overall PV system performance. This integration facilitates a comprehensive analysis of thermal dynamics and energy output, ultimately optimizing cooling strategies for greater efficiency under diverse operating conditions.1
- Advancing radiation models to achieve more realistic simulations of solar heating under dynamic environmental conditions. Incorporating such factors allows for more accurate predictions of thermal performance, thereby supporting the optimization of solar energy systems.1

Acknowledgements
Not applicable.

Authors' contributions
Idea for the article: Saleem Raza, Ik-Tae Im; Literature search: Saleem Raza, Ahsan Farooq, Muhammad Sohail Shezad, Wazahat Hussain, Zafar Abbas; Writing—original draft: Saleem Raza, Zafar Abbas; Writing—review and editing, Ik-Tae Im; Supervision, Ik-Tae Im.

Funding
The authors did not receive support from any organization for the submitted work.

Data availability
Inquiries can be directed to the corresponding author.

Declarations

Competing Interests
The authors have no competing interests to declare that are relevant to the content of this article.

Received: 2 September 2025 / Accepted: 25 December 2025
Published online: 07 January 2026

References

1. Tamer Nabil, T. M., & Mansour (2022). Augmenting the performance of photovoltaic panel by decreasing its temperature using various cooling techniques. *Results in Engineering*, *15*(May), 100564.
2. Dengfeng Du, J., Darkwa, G., & Kokogiannakis (2013). Thermal management systems for photovoltaics (PV) installations: A critical review. *Solar Energy*, *97*, 238–254.
3. Rehan, A. (2014). Effect of solar panel cooling on PV performance, Master Thesis, Southern Illinois University.
4. Leow, W., Zhe, M. I., Yusoff, A. A., Razak, M. I., Misrun, S., Ibrahim, M. I., Fahmi, A. S., & Rosmi (2019). Effect of water cooling temperature on photovoltaic panel performance by using computational fluid dynamics (CFD). *Journal of Advanced Research in Fluid Mechanics and Thermal Sciences*, *56*, 133–146.
5. Vimukthi Vithanage, S., Hosan, H., Perera, C., Galpaya, A., Induranga, A., Indupama, D., Wijesekara, K., & Koswattage (2025). Impact of water based nanofluids in heat exchanger type active solar PV cooling systems: A comparative CFD analysis. *International Journal of Science and Research Archive (IJSRA)*, 14(01), pp. 2582–8185. https://doi.org/10.30574/ijsra.2025.14.1.0148.
6. Mahdi, E. J., Algburi, S., Al-Abadi, N., Ahmed, O. K., & Ahmed, A. K. (2024). Photovoltaic panel cooling using ground source energy: CFD simulation. *Results in Engineering*, *22*, 102144.
7. Antara, I. G. F. B., Sucipta, M., Astawa, K., Wirawan, I. K. G., & Sukrawa, M. (2024). CFD simulation of photovoltaic thermal (PV/T) cooling system with various channel geometries. *Journal of Heat and Mass Transfer Research*, 11(2), pp. 297–306. https://doi.org/10.22075/jhmtr.2024.33787.1551.
8. Sardouei, M. M., Mortezapour, H., & Naeimi, K. J. (2018). Temperature distribution and efficiency assessment of different PVT water collector designs. *Sadhana,43*, Article 84. https://doi.org/10.1007/s12046-018-0826-x
9. Qi Zhou, P., Dong, M., Li, Z., & Wang (2023). Analyzing the interactions between photovoltaic system and its ambient environment using CFD techniques: A review. *Energy & Buildings*, *296*, 113394.
10. Nižetić, S., Giama, E., & Papadopoulos, A. M. (2018). Comprehensive analysis and general economic-environmental evaluation of cooling techniques for photovoltaic panels, part II: Active cooling techniques. *Energy Conversion and Management*, *155*, 301–323.
11. Tonui, J. K., & Tripanagnostopoulos, Y. (2007). Air-cooled PV/T solar collectors with low cost performance improvements. *Solar Energy*, *81*, 498–511.
12. Uzair Sajjad, M., Amer, H. M., Ali, A., Dahiya, N., & Abbas (2019). Cost effective cooling of photovoltaic modules to improve efficiency. *Case Studies in Thermal Engineering*, *14*, 100420.
13. Gökhan, & Ömeroğlu (2018). CFD analysis and electrical efficiency improvement of a hybrid PV/T panel cooled by forced air circulation. *International Journal of Photoenergy Vol*. https://doi.org/10.1155/2018/9139683. Article ID 9139683.
14. Abdullah Al Hasan, M., Abdullah, M., & Sazid, A. S. (2024). A computational analysis of air-cooled heat sinks designs for PV solar panel cooling with different fin numbers. *Heat Transfer,54*(2), 1462–1475. https://doi.org/10.1002/htj.23236
15. Malagouda Patil, A., Sidramappa, A. M., & Hebbale. , Vishwantha, J. S. (2024). Computational fluid dynamics (CFD) analysis of air-cooled solar photovoltaic (PV/T) panels, Materials today: PROCEEDINGS, vol. 100, pp. 93–101. https://doi.org/10.1016/j.matpr.2023.05.198.
16. Jian Sun, M., & Shi (2010). Experimental study on a concentrating solar photovoltaic/thermal system, *2010 Asia-Pacific Power and Energy Engineering Conference (APPEEC)*, pp. 2–5. https://doi.org/10.1109/APPEEC.2010.5449528
17. Artur Wodołazski, N., Howaniec, B., Jura, A., Bak, A., & smolinski (2021). CFD numerical modelling of a PV–TEG hybrid system cooled by air heat sink coupled with a single-phase inverter. *Materials*, *14*(9), 5800. https://doi.org/10.3390/ma14195800
18. Ahmet Numan Özakin, F., & Kaya (2019). Effect on the exergy of the PVT system of fins added to an air-cooled channel: A study on temperature and air velocity with ANSYS fluent. *Solar Energy*, *184*, 561–569.
19. Wei Liu, S., Lian, X., Fang, Z., Shang, H., Wu, H., Zhu, S., & Hu (2023). An open-source and experimentally guided CFD strategy for predicting air distribution in data centers with air-cooling. *Building and Environment*.
20. Moustafa, A. A., Al-Damook, P. D., & Thesis (2019). University of Leeds.
21. Ridha Hassan, M. H. K., Aboaltabooq, Z. A., & Jaafar (2020). Experimental and numerical study on the effect of water cooling on PV panel conversion efficiency, IOP Conference Series: *Materials Science and Engineering*, 928, 022094. https://doi.org/10.1088/1757-899X/928/2/022094,.
22. Sarthak Vaishnav, V., Kapadiya, R., Harish, R., & Mohan (2021). Design and analysis of energy-efficient solar panel cooling system using computational fluid dynamics, IOP Conference Series: *Materials Science and Engineering*, vol. 1128, 012033. https://doi.org/10.1088/1757-899X/1128/1/012033
23. Hrvoje Dedić-Jandrek, Ž., & Penga Computational fluid dynamics analysis of water spray cooling for photovoltaic panels, 2020 5th International Conference on Smart and Sustainable Technologies (SpliTech), Split, Croatia, 2020, pp. 1–6. https://doi.org/10.23919/SpliTech49282.2020.9243837

24. Wu, G., Liu, Q., & Wang, J. (2021). Thermal analysis of water-cooled photovoltaic cell by applying computational fluid dynamics. *Journal of Thermal Analysis and Calorimetry*, *144*, 1741–1747. https://doi.org/10.1007/s10973-020-10283-z
25. Pang, W., Cui, Y., Yu, H., Zhang, L., & Yan, H. (2019). Experimental effect of high mass flow rate and volume cooling on performance of a water-type PV/T collector. *Solar Energy*, *188*, 1360–1368.
26. Kazem, H. A., Al-Waeli, A. H. A., Chaichan, M. T., Al-Waeli, K. H., Al-Aasam, A. B., & Sopian, K. (2020). Evaluation and comparison of different flow configurations PVT systems in Oman: A numerical and experimental investigation. *Solar Energy,208*, 58–88.
27. Nowzari, R. (2019). Numerical analysis of a photovoltaic module integrated with various water cooling systems. *Transactions of FAMENA*. https://doi.org/10.21278/TOF.43202
28. Aswad, T. S. N., Sadiq Al-Baghdadi, M. A. R., Al-Waily, M., & Razali, M. A. B. (2021). Performance enhancement of a photovoltaic cell working in hot environment conditions using Al_2O_3 nanofluids: A CFD study. *International Journal of Nanoelectronics and Materials*, *14*, 317–328.
29. Rosli, M. A. M., Loon, Y. W., Nawam, M. Z., Misha, S., Roslizar, A., Hussain, F., Hamid, N. A., Arifin, Z., & Herawan, S. G. (2021). Validation study of photovoltaic thermal nanofluid based coolant using computational fluid dynamics approach. *CFD Letters*, *13*, 58–71. https://doi.org/10.37934/cfdl.13.3.5871
30. Jose, J. P. A., Shrivastava, A., Soni, P. K., Hemalatha, N., Alshahrani, S., Saleel, C. A., Sharma, A., Shaik, S., & Alarifi, I. M. (2023). An analysis of the effects of nanofluid-based serpentine tube cooling enhancement in solar photovoltaic cells for green cities. *Journal of Nanomaterials*. https://doi.org/10.1155/2023/3456536
31. Hasan, Husam Abdulrasool, Hatem, A. A., Abd, L. A., Abed, A. M., & Sopian, K. (2022). Numerical investigation of nanofluids comprising different metal oxide nanoparticles for cooling concentration photovoltaic thermal CPVT. *Cleaner Engineering and Technology,10*, Article 100543.
32. Omri, Mohamed, Selimefendigil, F., Smaoui, H. T., & Kolsi, L. (2022). Cooling system design for PVthermal management by using multiple porous deflectors and nanofluid. *Case Studies in Thermal Engineering,39*, Article 102405. https://doi.org/10.1016/j.csite.2022.102405
33. Shrivastava, Anurag, Jose, J. Prakash Arul., Borole, Yogini Dilip, Saravanakumar, R., Sharifpur, Mohsen, Harasi, Hossein, Razak, R.K. Abdul., & Afzal, Asif. (2022). A study on the effects of forced air-cooling enhancements on a 150 W solar photovoltaic thermal collector for green cities. *Sustainable Energy Technologies and Assessments,49*, Article 101782. https://doi.org/10.1016/j.seta.2021.101782
34. Ali, Hafiz M.. (2020). Recent advancements in PV cooling and efficiency enhancement integrating phase change materials based systems– A comprehensive review. *Solar Energy,197*, 163–198.
35. Nouira, Meriem, & Sammouda, Habib. (2018). Numerical study of an inclined photovoltaic system coupled with phase change material under various operating conditions. *Applied Thermal Engineering,141*, 958–975.
36. Hossam, Nermin, El-Hamid, M. Abd., Gamal, M., & Fatouh, M. (2025). Thermal management of photovoltaic modules using phase change materials: Experimental and numerical investigations. *Applied Thermal Engineering,279B*, Article 127645. https://doi.org/10.1016/j.applthermaleng.2025.127645
37. Aneli, Stefano, Arena, R., & Gagliano, A. (2021). Numerical simulations of a PV module with phase change material (PV-PCM) under variable weather conditions. *International Journal of Heat and Technology,39*, 643–652.
38. Dominic Groulx, P. H., Biwole, M., & Bhouri (2020). Phase change heat transfer in a rectangular enclosure as a function of inclination and fin placement. *International Journal of Thermal Sciences*, *151*, 106260.
39. Sumayli, Abdulrahman, Aashahrani, S. M., & Mahdi, W. A. (2023). The effect of the properties of porous fillers on the performance of a PCM-based photovoltaic thermal system: A CFD study. *Case Studies in Thermal Engineering,50*, Article 103444.
40. Khodadadi, Mohammadjavad, & Sheikholeslami, M. (2022). Assessment of photovoltaic thermal unit equipped with phase change material in different finned containers. *Journal of Energy Storage,46*, Article 103939. https://doi.org/10.1016/j.est.2021.103939
41. Wongwuttanasatian, T., Sarikarin, T., & Suksri, A. (2020). Performance enhancement of a photovoltaic module by passive cooling using phase change material in a finned container heat sink. *Solar Energy*, *195*, 47–53. https://doi.org/10.1016/j.solener.2019.11.053
42. Nada, S. A., El-Nagar, D. H., & Hussein, H. M. S. (2018). Improving the thermal regulation and efficiency enhancement of PCM-integrated PV modules using nano particles. *Energy Conversion and Management,166*, 735–743. https://doi.org/10.1016/j.enconman.2018.04.035
43. Kumar, K. Senthil., Kumar, H. Ashwin., Gowtham, P., Kumar, S. Hari Selva., & Sudhan, R. Hari. (2020). Experimental analysis and increasing the energy efficiency of PV cell with nano-PCM (calcium carbonate, silicon carbide, copper). *Materials Today: Proceedings,37*, 1221–1225. https://doi.org/10.1016/j.matpr.2020.06.430
44. Luo, Z., Huang, Z., Xie, N., Gao, X., Xu, T., Fang, Y., & Zhang, Z. (2017). Numerical and experimental study on temperature control of solar panels with form-stable paraffin/expanded graphite composite PCM. *Energy Conversion and Management*, *149*, 416–423. https://doi.org/10.1016/j.enconman.2017.07.046
45. Seto, Dewadono Bayu, Arifin, Zainal, Kristiawan, Budi, & Prasetyo, Singgih Dwi. (2024). Nanoparticle-enhanced phase change materials (NePCM) in passive cooling systems to improve solar panel efficiency. *International Review of Mechanical Engineering (IREME)*. https://doi.org/10.15866/ireme.v18i1.24149
46. NematpourKeshteli, Abolfazl, Iasiello, Marcello, Langella, Giuseppe, & Bianco, Nicola. (2024). Using metal foam and nanoparticle additives with different fin shapes for PCM-based thermal storage in flat plate solar collectors. *Thermal Science and Engineering Progress,52*, Article 102690. https://doi.org/10.1016/j.tsep.2024.102690
47. Sheikh, Yahya, Jasim, Muhammad, Qasim, Muhammad, Qaisieh, Alaa, Hamdan, Mohammad O.., & Abed, Farid. (2024). Enhancing PV solar panel efficiency through integration with a passive multi-layered PCMs cooling system: A numerical study. *International Journal of Thermofluids,23*, Article 100748. https://doi.org/10.1016/j.ijft.2024.100748
48. Alnakeeb, Mohamed A.., Abdel Salam, Mohamed A.., Hassab, Mohamed A.., & El-Maghlany, Wael M.. (2024). Numerical study of thermal and electrical performance of a new configuration of hybrid photovoltaic solar panel phase-change material cooling system. *Journal of Energy Storage,* Article 112945. https://doi.org/10.1016/j.est.2024.112945
49. Kim, Jaemin, & Nam, Yujin. (2019). Study on the cooling effect of attached fins on PV using CFD simulation. *Energies,12*(4), Article 758. https://doi.org/10.3390/en12040758
50. Pomares-Hernández, Cristhian, Zuluaga-García, Edwin Alexander, Escorcia Salas, Gene Elizabeth, Robles-Algarín, Carlos, & Sierra Ortega, Jose. (2021). Computational modeling of passive and active cooling methods to improve PV panels efficiency. *Applied Sciences,11*(23), Article 11370. https://doi.org/10.3390/app112311370
51. Beldjani, Charafeddine, Belghar, Nourredine, Aoues, Kamel, Saleh, Momen Sami Mohammed., Boutera, Yousra, & Kethiri, Mohamed Aymen. (2022). Efficiency improvement of air-cooled photovoltaic modules utilizing copper heat dissipators. *Desalination and Water Treatment,279*, 140–146. https://doi.org/10.5004/dwt.2022.29099
52. Komeili Birjandi, Amir, Eftekhari Yazdi, Mohammad, Dinarvand, Saeed, Salehi, Gholam Reza, & Tehrani, Pedram. (2021). Effect of using hybrid nanofluid in thermal management of photovoltaic panel in hot climates. *International Journal of Photoenergy*. https://doi.org/10.1155/2021/3167856
53. Potgieter, M. S. W., Bester, C. R., & Bhamjee, M. (2020). Experimental and CFD investigation of a hybrid solar air heater. *Solar Energy,195*, 413–428. https://doi.org/10.1016/j.solener.2019.11.058
54. Nasef, H. A., Nada, S. A., & Hassan, Hamdy. (2019). Integrative passive and active cooling system using PCM and nanofluid for thermal regulation of concentrated photovoltaic solar cells. *Energy Conversion and Management,199*, Article 112065.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.